D.K.T.E. Society's
Textile and Engineering Institute,
Rajwada, Ichalkaranji- 416115

(An Empowered Autonomous Institute Affiliated to Shivaji University, Kolhapur)

NAAC Accredited with A+ Grade, ISO 9001:2015 Certified

DEPARTMENT: Computer Science and Engineering
CURRICULUM
Third Year B.Tech. Engineering Program Under NEP2020

With effect from 2025-26

BoS Chairman | Dean Academics | Director

Teaching and Evaluation Scheme for Year 2025-26

Program: B. Tech. Computer Sc. & Engineering

Class.: Third Year, B. Tech. **Semester: SEM-V** **NCrF Level : 5.0** **W.E.F.: 2025-2026**

Sr. No.	Course Code	Course Title	Course Category	Teaching Scheme				Course Credits	Evaluation scheme					
									Theory			Practical		Total
									CIE		SEE	CIE	SEE	
				L	T	P	Contact Hrs/wk		SE-I	SE-II				
01	01CSL301	Database Engineering	PCC	3	-	-	3	3	25	25	50	-	-	100
02	01CSL302	Network Architecture and Protocols	PCC	3	-	-	3	3	25	25	50	-	-	100
03	PCE-I	Elective-I	PEC	3	-	-	3	3	25	25	50	-	-	100
04	01CSL303	Machine Learning	PCC	3	-	-	3	3	25	25	50	-	-	100
05	MDM-III	MDM-III	MDM	2	-	-	2	2	25	25	50	-	-	100
06	01CSP304	Machine Learning Lab	PCC	-	-	2	2	1	-	-	-	50	50	100
07	01CSP305	Database Engineering Lab	PCC	-	-	2	2	1	-	-	-	50	50	100
08	MDM-III Lab	MDM-III Lab	MDM	-	-	2	2	1	-	-	-	50	-	50
09	01CSPXX	Elective -I Lab	PEC	-	-	2	2	1	-	-	-	50	-	50
10	01CSD351	Software Mini Project	ELC	-	-	4	4	2	-	-	-	50	50	100
11	OE-II	Open Elective	OE	3	-	-	3	3	25	25	50	-	-	100
Total				**17**	**-**	**12**	**29**	**23**	**150**	**150**	**300**	**250**	**150**	**1000**

L- Lecture T-Tutorial P-Practical SE-I: Semester Examination-I SE-II: Semester Examination-II CIE – Continuous in Semester Evaluation SEE- Semester End Examination

	BSC/ESC		Program Courses		Multidisciplinary Courses		Skill Courses	Humanities Social Science and Management (HSSM)				Experiential Learning Courses				Liberal Learning Courses
Course Category	**BSC**	**ESC**	**PCC**	**PEC**	**MDM**	**OE**	**VSEC**	**AEC**	**Entrp / Mgmt**	**IKS**	**VEC**	**RM**	**CEP/FP**	**Proj**	**Int/ OJT**	**CC**
Credits	-	-	11	4	3	3	-	-	-	-	-	-	-	2	-	-
Cum. Sum	14	13	41	4	9	6	7	5	4	2	4	-	-	4	-	2

Progressive Total Credits: 92+23=115

Teaching and Evaluation Scheme for Year 2025-26

Program: B. Tech. Computer Sc. & Engineering

Class.: Third Year, B. Tech. Semester: SEM-VI NCrF Level : 5.0 W.E.F.: 2025-2026

Sr. No.	Course Code	Course Title	Course Category	Teaching scheme				Course Credits	Evaluation scheme					
									Theory			Practical		Total
									CIE		SEE	CIE	SEE	
				L	T	P	Contact Hrs/wk		SE-I	SE-II				
01	01CSL306	Cryptography and Network Security	PCC	3	-	-	3	3	25	25	50	-	-	100
02	PEC-II	Elective-II	PEC	3	-	-	3	3	25	25	50	-	-	100
03	PEC-III	Elective III	PEC	3	-	-	3	3	25	25	50	-	-	100
04	MDM-IV	MDM-IV	MDM	2	-	-	2	2	25	25	50	-	-	100
05	01CSP357	Robotics Process Automation	VSEC	-	-	2	2	1	-	-	-	50	50	100
06	PEC-II Lab	Elective-II Lab	PEC	-	-	2	2	1	-	-	-	50	-	50
07	PEC-III Lab	Elective III Lab	PEC	-	-	2	2	1	-	-	-	50	-	50
08	MDM-IV Lab	MDM-IV Lab	MDM	-	-	2	2	1	-	-	-	50	-	50
09	01CSP307	Full Stack Web Development	PCC	2		2	4	3	-	-	-	50	50	100
10	01CSD352	Project-I	CEP	-	-	2	2	2	-	-	-	50	50	100
11	OE-III	Open Elective	OE	2	-	-	2	2	25	25	50	-	-	100
12	01CST353	Internship	CEP	-	-	-	-	2				50		50
Total				**15**	**-**	**12**	**27**	**24**	**125**	**125**	**250**	**350**	**150**	**1000**

L- Lecture T-Tutorial P-Practical SE-I: Semester Examination-I SE-II: Semester Examination-II CIE – Continuous in Semester Evaluation SEE- Semester End Examination

	BSC/ESC		Program Courses		Multidisciplinary Courses		Skill Courses	Humanities Social Science and Management (HSSM)				Experiential Learning Courses				Liberal Learning Courses
Course Category	**BSC**	**ESC**	**PCC**	**PEC**	**MDM**	**OE**	**VSEC**	**AEC**	**Entrp / Mgmt**	**IKS**	**VEC**	**RM**	**CEP/FP**	**Proj**	**Int/ OJT**	**CC**
Credits	**-**	**-**	**6**	**8**	**3**	**2**	**1**	**-**	**-**	**-**	**-**	**-**	**2**	**-**	**2**	**-**
Cum. Sum	**14**	**13**	**48**	**12**	**12**	**8**	**8**	**4**	**4**	**2**	**4**	**-**	**2**	**4**	**2**	**2**

Progressive Total Credits: 115+24=139

Open Electives:

Sr. No.	Course Code	Course Name
1	01OEL301	Digital Marketing
2	01OEL302	Total Quality Management
3	01OEL303	Startup and Entrepreneurship
4	01OEL304	Project Management
5	01OEL305	Human Resource Management
6	01OEL351	German Language
7	01OEL352	Industrial Management
8	01OEL358	Professional Skills for Industry Readiness

List of Programme Elective Courses (PEC)

Elective No	Sem	Cloud Computing		Artificial Intelligence		Data Analytics	
		Course Code	Course Title	Course Code	Course Title	Course Code	Course Title
Elective-I	V	01CSL331	Introduction to Cloud Computing and Cloud Services	01CSL332	Deep Learning	01CSL333	Foundations of Data Science
Elective-I Lab	V	01CSP334	Introduction to Cloud Computing and Cloud Services Lab	01CSP335	Deep Learning Lab	01CSP336	Foundations of Data Science Lab
Elective-II	VI	01CSL337	Networking and Security in Cloud	01CSL338	Generative AI	01CSL339	Data Warehousing and Business Intelligence
Elective-II Lab	VI	01CSP340	Networking and Security in Cloud Lab	01CSP341	Generative AI Lab	01CSP342	Data Warehousing and Business Intelligence Lab
Elective-III	VI	01CSL343	Cloud Database and Analytics Services	01CSL344	Large Language Models	01CSL345	Data Analytics
Elective-III Lab	VI	01CSP346	Cloud Database and Analytics Services Lab	01CSP347	Large Language Models Lab	01CSP348	Data Analytics Lab
Elective-IV	VII	01CSL431	DevOps and Automation Tools	01CSL432	Agentic AI	01CSL433	Data Wrangling in Data Science

Elective No	Sem	Cloud Computing		Artificial Intelligence		Data Analytics	
		Course Code	Course Title	Course Code	Course Title	Course Code	Course Title
Elective-IV Lab	VII	01CSP434	DevOps and Automation Tools Lab	01CSP435	Agentic AI Lab	01CSP436	Data Wrangling in Data Science Lab
Elective-V	VIII	01CSL437	Cloud-Native Application Development	01CSL438	Prompt Engineering, RAG & LangChain	01CSL439	Time series analysis

Multidisciplinary Minor (MDM)

- Students should select any one minor, excluding those offered by their parent department, from the diverse range of minors available.
- Students should complete all courses within their selected minor in order to earn credits.

MDM Name	Sr. No	Course Code	Course Name	Sem	Offered by Department
Fabric Manufacturing Technology	1	01TTMDL1201	General Textile Technology	III	Textile Technology
	2	01TTMDP1202	General Textile Technology - Lab	III	
	3	01TTMDL1203	Woven Fabric Technology	IV	
	4	01TTMDP1204	Woven Fabric Technology - Lab	IV	
	5	01TTMDL1301	Knitted Fabric Technology	V	
	6	01TTMDL1302	Knitted Fabric Technology - Lab	V	
	7	01TTMDL1303	Nonwoven Fabric Technology	VI	
	8	01TTMDL1304	Nonwoven Fabric Technology - Lab	VI	
	9	01TTMDD1401	Capstone Project	VII	
High Performance Textiles	1	01TMMDL1201	Fundamentals of High Performance Textiles	III	Man Made Textile Technology
	2	01TMMDP1202	Fundamentals of High Performance Textiles Lab	III	
	3	01TMMDL1203	High Performance Fibres	IV	
	4	01TMMDP1204	High Performance Fibres Lab	IV	
	5	01TMMDL1301	Manufacturing of High Performance Textiles	V	
	6	01TMMDP1302	Manufacturing of High Performance Textiles Lab	V	
	7	01TMMDL1303	Automotive and Geo Textiles	VI	
	8	01TMMDP1304	Automotive and Geo Textiles Lab	VI	
	9	01TMMDD1401	Capstone Project	VII	
Machine Maintenance	1	01MEMDL1201	Applied Electronics	III	Textile Plant Engineering
	2	01MEMDP1202	Applied Electronics Lab	III	
	3	01MEMDL1203	Computer Programing	IV	
	4	01MEMDP1204	Computer Programing Lab	IV	
	5	01MEMDL1301	Fluid Flow Systems and Control	V	
	6	01MEMDL1302	Fluid Flow Systems and Control Lab	V	
	7	01MEMDL1303	Mechatronics	VI	
	8	01MEMDT1304	Mechatronics Lab	VI	

MDM Name	Sr. No	Course Code	Course Name	Sem	Offered by Department
	9	01MEMDD1401	Capstone Project	VII	
Chemical Processing of Textiles	1	01TCMDL1201	Introduction to Textile Substrates	III	Textile Chemistry
	2	01TCMDP1202	Introduction to Textile Substrates Lab	III	
	3	01TCMDL1203	Pretreatment of Textiles	IV	
	4	01TCMDP1204	Pretreatment of Textiles Lab	IV	
	5	01TCMDL1301	Colouration of Textiles	V	
	6	01TCMDP1302	Colouration of Textiles Lab	V	
	7	01TCMDL1303	Finishing & Care of Textiles	VI	
	8	01TCMDP1304	Finishing & Care Textiles Lab	VI	
	9	01TCMDD1401	Capstone Project	VII	
Garment Technology	1	01TFMDL1201	Fundamentals of Clothing Manufacturing	III	Fashion Technology
	2	01TFMDP1202	Fundamentals of Clothing Manufacturing Lab	III	
	3	01TFMDL1203	Garment Manufacturing	IV	
	4	01TFMDP1204	Garment Manufacturing Lab	IV	
	5	01TFMDL1301	Design & Pattern Engineering	V	
	6	01TFMDP1302	Design & Pattern Engineering Lab	V	
	7	01TFMDL1303	Fashion and Garment construction	VI	
	8	01TFMDP1304	Fashion and Garment construction Lab	VI	
	9	01TFMDD1401	Capstone Project	VII	
Embedded Systems	1	01ETMDL1201	Microcontroller	III	Electronics and Telecommunication
	2	01ETMDP1202	Microcontroller Lab	III	
	3	01ETMDL1203	Microcontroller based System	IV	
	4	01ETMDP1204	Microcontroller based System Lab	IV	
	5	01ETMDL1301	Embedded System Design	V	
	6	01ETMDP1302	Embedded System Design Lab	V	
	7	01ETMDL1303	Real Time Embedded System	VI	
	8	01ETMDP1304	Real Time Embedded System Lab	VI	
	9	01ETMDD1401	Capstone Project	VII	
Industrial IOT	1	01ETMDL2201	Sensor and Actuators	III	Electronics and Telecommunication
	2	01ETMDP2202	Sensor and Actuators Lab	III	
	3	01ETMDL2203	Introduction to IOT	IV	
	4	01ETMDP2204	Introduction to IOT Lab	IV	
	5	01ETMDL2301	IOT implementation using Raspberry Pi Boards	V	
	6	01ETMDP2302	IOT implementation using Raspberry Pi Boards Lab	V	
	7	01ETMDL2303	IOT System design	VI	
	8	01ETMDP2304	IOT System design Lab	VI	
	9	01ETMDD2401	Capstone Project	VII	
Industrial Electronics	1	01ETMDL3201	Digital Electronics	III	Electronics and Telecommunication
	2	01ETMDP3202	Digital Electronics Lab	III	
	3	01ETMDL3203	Control System	IV	
	4	01ETMDP3204	Control System Lab	IV	
	5	01ETMDL3301	Microcontroller	V	
	6	01ETMDP3302	Microcontroller Lab	V	
	7	01ETMDL3303	Power Electronics and Drives	VI	
	8	01ETMDP3304	Power Electronics and Drives Lab	VI	
	9	01ETMDD4401	Capstone Project	VII	
Digital System Engineering	1	01ETMDL4201	Basics of Electronics	III	Electronics and Telecommunication
	2	01ETMDP4202	Basics of Electronics Lab	III	
	3	01ETMDL4203	Computer Programming	IV	

MDM Name	Sr. No	Course Code	Course Name	Sem	Offered by Department
	4	01ETMDP4204	Computer Programming Lab	IV	
	5	01ETMDL4301	Applied Electronics and SAP	V	
	6	01ETMDP4302	Applied Electronics and SAP Lab	V	
	7	01ETMDL4303	AI and Data Science	VI	
	8	01ETMDP4304	AI and Data Science Lab	VI	
	9	01ETMDD4401	Capstone Project	VII	
Industrial Automation	1	01MEMDL2201	Introduction to Automation	III	Mechanical
	2	01MEMDP2202	Introduction to Automation Lab	III	
	3	01MEMDL2203	Mechatronics	IV	
	4	01MEMDP2204	Mechatronics Lab	IV	
	5	01MEMDL2301	Industrial Robotics	V	
	6	01MEMDP2302	Industrial Robotics Lab	V	
	7	01MEMDL2303	Control Systems	VI	
	8	01MEMDP2304	Control Systems Lab	VI	
	9	01MEMDP2401	Capstone Project	VII	
Product Design and Development	1	01MEMDL1201	Introduction to Product Design	III	Mechanical
	2	01MEMDP1202	Introduction to Product Design Lab	III	
	3	01MEMDL1203	Materials and Manufacturing Processes for Product Design	IV	
	4	01MEMDP1204	Materials and Manufacturing Processes for Product Design Lab	IV	
	5	01MEMDL1301	CAD and Digital Prototyping	V	
	6	01MEMDP1302	CAD and Digital Prototyping Lab	V	
	7	01MEMDL1303	Product Lifecycle Management	VI	
	8	01MEMDP1304	Product Lifecycle Management Lab	VI	
	9	01MEMDP1401	Capstone Project	VII	
Energy Engineering	1	01MEMDL3201	Thermal Science	III	Mechanical
	2	01MEMDP3202	Thermal Science Lab	III	
	3	01MEMDL3203	Heat Transfer	IV	
	4	01MEMDP3204	Heat Transfer Lab	IV	
	5	01MEMDL3301	Chemical Engineering Operations	V	
	6	01MEMDP3302	Chemical Engineering Operations Lab	V	
	7	01MEMDL3303	Fluid Mechanics and Machines	VI	
	8	01MEMDP3304	Fluid Mechanics and Machines Lab	VI	
	9	01MEMDD3401	Capstone Project	VII	
Automation and Robotics	1	01MEMDL4201	Applied Electronics	III	Mechanical
	2	01MEMDP4202	Applied Electronics Lab	III	
	3	01MEMDL4203	Computer Programming	IV	
	4	01MEMDP4204	Computer Programming Lab	IV	
	5	01MEMDL4301	Fluid Flow Systems and Control	V	
	6	01MEMDP4302	Fluid Flow Systems and Control Lab	V	
	7	01MEMDL4303	Mechatronics and Robotics	VI	
	8	01MEMDP4304	Mechatronics and Robotics Lab	VI	
	9	01MEMDD4401	Capstone Project	VII	
Software Engineering	1	01CSMDL1201	Software Engineering Processes	III	Computer Science and Engineering
	2	01CSMDL1202	Object Oriented Design and Programming	IV	
	3	01CSMDP1203	Java Programming	IV	
	4	01CSMDL1301	Database Engineering	V	
	5	01CSMDP1302	Database Engineering Lab	V	
	6	01CSMDL1303	Software Testing	VI	
	7	01CSMDP1304	Software Testing Lab	VI	
	8	01CSMDD1401	Capstone Project	VII	
	1	01CSMDL2201	Fundamentals of Computer Networks and Database Systems	III	

MDM Name	Sr. No	Course Code	Course Name	Sem	Offered by Department
Web Technology	2	01CSMDP2202	Computer Networks and Database Systems Lab	III	
	3	01CSMDP2203	Web Page Designing Lab	IV	
	4	01CSMDP2301	Client-Side Programming Lab	V	
	5	01CSMDP2302	Server Side Programming Lab	VI	
	6	01CSMDD2401	Capstone Project	VII	
Digital Design and Manufacturing Systems	1	01CSMDL3201	Basics of Electronic Systems	III	
	2	01CSMDP3202	Basics of Electronic Systems Lab	III	
	3	01CSMDL3203	Digital Illustration Techniques	IV	
	4	01CSMDP3204	Digital Illustration Techniques Lab	IV	
	5	01CSMDL3301	Programming techniques and applications	V	
	6	01CSMDP3302	Programming techniques and applications Lab	V	
	7	01CSMDL3303	CAD CAM Applications	VI	
	8	01CSMDP3304	CAD CAM Applications Lab	VI	
	9	01CSMDD3401	Capstone Project	VII	
Artificial Intelligence and Machine Learning	1	01AMMDL1201	Introduction to AI-ML	III	Computer Science and Engineering (Artificial Intelligence and Machine Learning)
	2	01AMMDP1202	Basics of AI-ML Lab	III	
	3	01AMMDL1203	Programming with Python for Machine Learning	IV	
	4	01AMMDL1301	Machine Learning Foundations	V	
	5	01AMMDP1302	Machine Learning Foundations Lab	V	
	6	01AMMDL1303	Ethics and Applications in AI	VI	
	7	01AMMDP1304	Deep Learning Lab	VI	
	8	01AMMDD1401	Capstone Project on AI-ML	VII	
Augmented Reality and Virtual Reality	1	01AMMDL2201	Foundations of AR-VR	III	
	2	1AMMDP2202	Basic AR-VR Lab	III	
	3	01AMMDL2203	Programming for AR-VR and 3D fundamentals	IV	
	4	01AMMDP2204	Programming for AR-VR and 3D fundamentals	IV	
	5	01AMMDL2301	3D Modeling and Animation	V	
	6	01AMMDL2302	Virtual Reality Development	VI	
	7	01AMMDD2401	Capstone Project	VII	
Data Analytics	1	01ADMDL1201	Foundation of Data Analysis	III	Artificial Intelligence and Data Science
	2	01ADMDP1202	Foundation of Data Analysis Lab	III	
	3	01ADMDL1203	Exploratory Data Analysis	IV	
	4	01ADMDP1204	Exploratory Data Analysis Lab	IV	
	5	01ADMDL1301	Data Analysis Techniques	V	
	6	01ADMDP1302	Data Analysis Techniques Lab	V	
	7	01ADMDL1303	Data Visualization Techniques	VI	
	8	01ADMDP1304	Data Visualization Techniques Lab	VI	
	9	01ADMDP1401	Capstone Project	VII	
Construction Technology	1	01CEMDL1201	Building Materials & Construction	III	Civil
	2	01CEMDP1202	Building Materials & Construction Lab	III	
	3	01CEMDL1203	Metrics in Civil Engineering	IV	
	4	01CEMDP1204	Metrics in Civil Engineering Lab	IV	
	5	01CEMDL1305	Estimation & Valuation	V	
	6	01CEMDP1306	Estimation & Valuation Lab	V	
	7	01CEMDL1307	Project Planning & Management	VI	
	8	01CEMDP1308	Project Planning & Management Lab	VI	
	9	01CEMDD1409	Capstone Project	VII	

MDM Name	Sr. No	Course Code	Course Name	Sem	Offered by Department
Industrial Electrical System	1	01EEMDL12	Electrical Installation System	III	Electrical
	2	01EEMDP1202	Electrical Installation System Lab	III	
	3	01EEMDL1203	Electrical Switchgears and Safety	IV	
	4	01EEMDP1204	Electrical Switchgears and Safety Lab	IV	
	5	01EEMDL1305	Renewable Energy and Cogeneration in Industry	V	
	6	01EEMDP1306	Renewable Energy and Cogeneration in Industry Lab	V	
	7	01EEMDL1307	Electrical Vehicle Technology	VI	
	8	01EEMDP1308	Electrical Vehicle Technology Lab	VI	
	9	01EEMDD1409	Capstone Project	VII	

Third Year B. Tech. (CSE) (Semester–V)							
Course Code	**01CSL301**	**Course Name**	**Database Engineering**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3	0	0	3		25	25	50

Pre-requisites: Data Structure

Course Objectives:

01	Introduce the foundational concepts and architecture of database management systems (DBMS).
02	Enable students to analyze and model real-world systems using E-R and Enhanced E-R models.
03	Teach structured query language (SQL) and Equip students with knowledge of PL/SQL for procedural database programming.
04	Provide an understanding of transaction processing, concurrency control, and database recovery.

Course Outcomes:

At the end of the course, students will be able to

01	Explain fundamental DBMS concepts and analyze transaction processing, concurrency control, and recovery mechanisms..
02	Design normalized relational database schemas using ER modeling and functional dependencies.
03	Develop optimized SQL queries and PL/SQL programs using advanced constructs like joins, subqueries, triggers, procedures, and analyze query execution strategies

Course Contents

Unit I	Introduction to Database Systems	7 Hours

Introduction: Purpose and Benefits of Database Systems, Characteristics of DBMS vs File System, Database Users and Administrators, DBMS Architecture, Data Abstraction and Independence, Database Schemas, Keys, Data Models: Relational, Hierarchical, Network, Object-based, Semi-structured, Database Languages: DDL, DML, DCL, TCL, Relational Algebra.

Unit II	Database Design and Normalization	6 Hours

Database Design Process, ER Model: Entities, Relationships, Attributes, ER Diagrams and EER Concepts (ISA, Aggregation), Mapping ER to Relational Schema, Integrity Constraints, Functional Dependencies, Normal Forms: 1NF, 2NF, 3NF, BCNF, Multivalued Dependencies and 4NF, Decomposition, Dependency Preservation.

Unit III	Structured Query Language (SQL)	7 Hours

SQL Basics: Syntax and Structure, Data Definition Language (DDL): CREATE, ALTER, DROP, Data Manipulation Language (DML): SELECT, INSERT, UPDATE, DELETE, SQL Clauses: WHERE, GROUP BY, HAVING, ORDER BY, JOIN Operations: INNER, OUTER (LEFT, RIGHT, FULL), SELF, CROSS, Set Operations: UNION, INTERSECT, EXCEPT, Subqueries: Nested and Correlated, Views, Indexes, and Sequences, Aggregate Functions and Conditional Expressions (CASE, COALESCE).

Unit IV	PL/SQL – Procedural SQL (Extended from SQL Concepts)	6 Hours

Basics of PL/SQL and Program Structure, Variables, Constants, and Data Types, Control Structures: IF, LOOP, WHILE, FOR, Cursors: Implicit and Explicit, Exception Handling, Stored Procedures and Functions, Triggers, Packages and Modular PL/SQL Programming

Unit V	Transactions, Concurrency Control, and Recovery	6 Hours

Transactions: Definition and Properties (ACID), Transaction States and Lifecycle, Schedules, Serializability-Conflict & View, Concurrency Control Techniques- Lock-Based Protocols, Two-Phase Locking (2PL) and Its Variations, Introduction of Timestamp-Based Protocols, Need for Recovery, Types of Failures, Recovery Techniques: Log-Based Recovery (Undo, Redo, and Undo–Redo), Checkpoints, Deferred and Immediate Update Methods.		
Unit VI	**Query Processing and Optimization**	**5 Hours**
Phases of query processing, Query execution plan, Importance of cost estimation, Sequential scan vs. index scan, Join algorithms, Query Optimization Techniques		
Texts Books:		
1.	Database System Concepts – Abraham Silberschatz, Henry Korth, S. Sudarshan (7th Edition or latest)	
2.	"Fundamentals of Database Systems" – Elmasri & Navathe	
3.	"Database Systems - A Practical Approach to Design, Implementation and Management", Thomas Connolly, Carolyn Begg, 4th Edition, Addison Wesley.	
Reference Books:		
1	"SQL, PL/SQL – The Programming Language of Oracle" – Ivan Bayross	
2	"Oracle PL/SQL Programming" – Steven Feuerstein	
Supplementary Readings:		
1. Oracle PL/SQL Programming – Steven Feuerstein 2. Learning SQL – Alan Beaulieu 3. Online Documentation (PostgreSQL, Oracle, MySQL)		

Third Year B. Tech. (CSE) (Semester–V)							
Course Code	**01CSL302**	**Course Name**	**Network Architecture and Protocols**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3	0	0	3		25	25	50

Pre-requisites: Introduction to computer network

Course Objectives:

01	Study of network architectures
02	Study of different issues and their solutions
03	Study of protocols used in network
04	Study of reliability of data transmission

Course Outcomes:

At the end of the course, students will be able to

01	Describe the basic concepts in Computer Network
02	Apply network concepts for real life problem.
03	Analyse different methods and protocols used in computer network

Course Contents

Unit I	Introduction to Computer Network	6 Hours

Applications of computer network, types of network, layered network architecture, design issues for the layers, connection oriented and connection less service, service primitives, service primitives and protocols. Reference models, architecture of internet, wireless and sensor network.

Unit II	Data Link Layer	6 Hours

Introduction to physical layer and guided media, Computer-to -Computer communication, data link layer design issues, service provided to network layer, Framming, Error Control-error detection and correction, elementary datalink protocols- simplex, stop and wait, and sliding window protocols

Unit III	Medium Access Control Sublayer	6 Hours

Need of MAC sublayer, channel allocation problem, static and dynamic channel allocation, multiple access protocols- ALOHA, Carrier Sense Multiple Access protocols, Collision free protocols, Ethernet – architecture, MAC sublayer protocol, switched ethernet, Ethernet standards, RFID

Unit IV	Network Layer	8 Hours

Need and design issues, service provided to transport layer, implementation of connection oriented and connection-less service. Routing algorithms - shortest path, flooding, distance vector routing, link state routing, Broadcast routing. Internetworking, Tunneling, IPv4 protocol-addressing, subnets, Classful and Special Addressing, Classless addressing, NAT, ARP

Unit V	Transport Layer	8 Hours

Transport service, service primitives, Elements of transport protocol-addressing, connection establishment, connection release, Internet Transport Protocol -UDP, Remote Procedure call. TCP Protocol-TCP service model, TCP connection establishment and release. TCP sliding window. Performance problems in Computer Network.

Unit VI	Application Layer	4 Hours

Domain Name System – namespace, name record, name server. World Wide Web- architecture, client and server side, HTTP protocol-connection, methods, headers.	
Texts Books:	
1.	Computer Networks by Andrew S. Tanenbaum, 5th Edition, Pearson publication
Reference Books:	
1	Data Communications and Networking with TCPIP Protocol Suite by Behrouz A. Forouzan 6th Edition, McGraw Hill publication
Supplementary Readings:	
1.	Computer Networking: A Top-Down Approach by James F. Kurose & Keith W. Ross, 6th edition, Pearson

Third Year B. Tech. (CSE) (Semester–V)							
Course Code	**01CSL331**		**Course Name**	**Introduction to Cloud Computing and Cloud Services**			
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3	0	0	3		25	25	50

Course Objectives	
01	Introduce fundamentals of AWS Lambda and Amazon S3
02	Understand working of Amazon EBS and EFS
03	Understand storage and snowball
Course Outcomes:	
At the end of the course, students will be able to	
01	Describe the basic concepts of compute, storage and hosting services in the AWS cloud.
02	Apply compute, storage and hosting services in real life scenarios.
03	Compare and contrast different AWS storage services (S3, EBS, EFS) and analyze their appropriate use in various application scenarios.

Course Contents		
Unit I	**Fundamentals of Cloud Computing and Services**	**6 Hours**
Basics of Cloud Computing – IaaS, PaaS, SaaS, Deployment Models, AWS Global Infrastructure – Regions, Availability Zones, Edge Locations, Identity and Access Management (IAM), Introduction to AWS CLI and SDKs Introduction to Amazon EC2, Launch and Connect to EC2 Instances, Configuring EC2 Instances, EC2 Storage and Backup, Auto Scaling in EC2, Load Balancing EC2 Instances		
Unit II	**AWS Lambda – Serverless Functions and Use Cases**	**6 Hours**
Introduction to Serverless Computing and AWS Lambda, AWS Lambda Architecture and Core Concepts, Writing and Deploying Lambda Functions, Integrating AWS Lambda with Other AWS Services, Advanced Lambda Features and Deployment Tools, Real-World Use Cases and Optimization Techniques		
Unit III	**Amazon S3 – Buckets, Policies, Versioning, Lifecycle**	**6 Hours**
Introduction to Amazon S3 & Buckets, S3 Bucket Policies and Access Management, Fine-Grained Access Control with IAM & S3 Permissions, S3 Versioning, Lifecycle Management in Amazon S3, Best Practices, Monitoring & Hands-on Use Case.		
Unit IV	**Amazon EBS and EFS – Block vs File Storage**	**6 Hours**
Introduction to AWS Storage Services, Amazon Elastic Block Store (EBS) – Concepts and Architecture, Amazon Elastic File System (EFS) – Concepts and Architecture, Comparison: Block Storage (EBS) vs. File Storage (EFS), Best Practices and Performance Optimization, Hands-on and Real-World Use Cases		
Unit V	**Amazon Lightsail – Simplified Cloud Hosting**	**6 Hours**
Introduction to Amazon Lightsail, Lightsail Instances and Configuration, Networking and Storage in Lightsail, Lightsail Databases and Containers, Scaling and Monitoring, Integration with Other AWS Services & Use Cases.		
Unit VI	**Storage Gateway and Snowball – Hybrid and Data Transfer Solutions**	**6 Hours**

Introduction to Hybrid Cloud Storage and AWS Data Transfer Services, AWS Storage Gateway – Architecture and Types, AWS Storage Gateway – Use Cases and Management, AWS Snowball – Overview and Use Cases, Snowball Architecture, Workflow and Security, Comparison, Integration, and Best Practices	
Texts Books:	
1.	AWS Certified Solutions Architect Study Guide: Associate (SAA-C03) Exam By: Ben Piper, David Clinton
Reference Books:	
1	Learning Amazon Web Services (AWS): A Hands-On Guide to the Fundamentals of AWS Cloud By: Mark Wilkins
2	Amazon Web Services in Action By: Michael Wittig & Andreas Wittig
Supplementary Readings:	
https://docs.aws.amazon.com/whitepapers/latest/aws-overview/introduction.html	

Third Year B. Tech. (CSE) (Semester–V)							
Course Code	**01CSL332**		**Course Name**	**Deep Learning**			
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3	0	0	3		25	25	50

Pre-requisites:

Course Objectives:

01	Teach the foundational principles and components of deep neural networks, including their architecture, activation functions, and learning mechanisms.
02	Understand and explain the structure, functionality, and differences between various deep learning models such as CNNs, RNNs, and LSTMs.
03	Teach students how to evaluate and compare deep learning models based on theoretical parameters like accuracy, convergence behavior, generalization, and complexity.

Course Outcomes:

At the end of the course, students will be able to

01	Explain concepts of deep neural network.
02	Apply functionality of different deep learning models to the given problem
03	Analyze deep learning architectures using performance metrics and design considerations.

Course Contents

Unit I	Introduction to Deep Learning	6 Hours

Relation between artificial intelligence, machine learning, deep learning, Introduction to neural network, Handling non-linear patterns, Activation functions, Artificial Neural Network, Working of ANN with forward pass and backword propagation, Loss Functions: MSE, MAE, Huber Loss, Binary Cross Entropy, Categorical Cross Entropy, Sparse Categorical Cross Entropy , Computational Graph and partial differentiation, Optimization, Gradient Descent Algorithm, Learning Rate, Effect of Learning Rate.

Unit II	Working with Deep Neural Network	6 Hours

Tensorflow, Tensors, Keras, Hello World Program: Classify handwritten digits, Improving your model, Distribution of dataset for improving accuracy of model, Hyperparameter tunning, Bias and variance of model: Underfitting, Overfitting and good fitting, Handling high variance: Adding more data, L1 and L2 regularization, Drop out, Early stopping, Handling high bias, Batch normalization, Learning rate decay, Advanced Optimizers: Gradient Descent with Momentum, Saving your model and deployment.

Unit III	Convolutional Neural Network	6 Hours

Challenges with larger image, Convolution Operation, Convolution example, Identifying features with convolution, Standard Convolution Filters, Creating Convolution Layer, Properties of Convolution: Translation Invariant, Spatial Hierarchical of Patterns, Stride, Pooling, Padding, Channel, Complete CNN model, ConvNet Example: Classifying image of objects, Working with your own dataset, Using Pre-trained ConvNet, Fine tunning, feature extraction, Visualizing feature map and filters.

Unit IV	Advanced Convolutional Neural Network	6 Hours

Popular CNN Architectures: LeNet-5, AlexNet, VGGNet, ResNet with Skip Connections, MobileNet, Advanced CNN Concepts: 1x1 Convolution, InceptionNet with 1X1 Convolution, Transpose Convolution,

Application of CNN: Object Detection, YOLO Model for Object Detection, Semantic Segmentation, Types of Semantic Segmentation, Fully Convolutional Network (FCN) for Image Segmentation,		
Unit V	**Sequential Models**	**6 Hours**
Sequential Data, Working with text data, Text encoding: One hot encoding, Word embedding, Embedding model: CBOW, Skip-gram Models, Learning word embeddings with main task, Using pre-trained word embedding, Understanding recurrent neural networks, A recurrent layer in Tensorflow, IMDB movie classification using SimpleRNN, Gated Recurrent Units (GRU), Long-short term memory (LSTM), Bidirectional RNN.		
Unit VI	**Advanced Deep Learning Best Practices**	**6 Hours**
Types of Models: One-to-One, One-to-Many, Many-to-One, Many-to-Many, Functional API, Layer weight sharing, Callbacks, Getting Most out of your model: Normalization and Standardization of data, Hyper-parameter optimization, Model ensemble, Speed up training on GPU with mixed precision, Multi-GPU training, TPU training.		
Texts Books:		
1.	Deep Learning by Ian Goodfellow, Yoshua Bengio, Aaron Courville, MIT Press Book	
2.	Deep Learning with Python by Francois Chollet, Manning Publications	
Reference Books:		
1	Deep Learning: A Practitioner's Approach by Josh Patterson and Adam Gibson	
Supplementary Readings:		
1. https://www.deeplearning.ai/courses/tensorflow-advanced-techniques-specialization/		

Third Year B. Tech. (CSE) (Semester–V)							
Course Code	**01CSL333**		**Course Name**	**Foundations of Data Science**			
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3	0	0	3		25	25	50

Pre-requisites:

Course Objectives:

01	To provide a comprehensive knowledge of data science using Python.
02	To learn the essential concepts of data analytics and data visualization.

Course Outcomes:

At the end of the course, students will be able to

01	Apply principles of NumPy and Pandas to the analysis of data
02	Make use of various file formats in loading and storage of data.
03	Identify and apply the need and importance of pre-processing techniques.
04	Show the results and present them in a pictorial format.

Course Contents

Unit I	Introduction to Data Science	6 Hours

Data science: definition, Datafication, Exploratory Data Analysis, The Data science process, A data scientist role in this process.

NumPy Basics: The NumPy ndarray: A Multidimensional Array Object, creating ndarrays, Data Types for ndarrays, Operations between Arrays and Scalars, Basic Indexing and Slicing, Boolean Indexing, Fancy Indexing, Data Processing Using Arrays, Expressing Conditional Logic as Array Operations, Methods for Boolean Arrays, Sorting, Unique

Unit II	Getting Started with pandas	6 Hours

Introduction to pandas, Library Architecture, Features, Applications, Data Structures, Series, DataFrame, Index Objects, Essential Functionality Reindexing, dropping entries from an axis, Indexing, selection, and filtering), Sorting and ranking, Summarizing and Computing Descriptive Statistics, Unique Values, Value Counts, Handling Missing Data, filtering out missing data.

Unit III	Data Loading, Storage, and File Formats	6 Hours

Reading and Writing Data in Text Format, Reading Text Files in Pieces, Writing Data Out to Text Format, Manually Working with Delimited Formats, JSON Data, XML and HTML: Web Scraping, Binary Data Formats, Using HDF5 Format, Reading Microsoft Excel Files, Interacting with Databases, Storing and Loading Data in MongoDB of EDA – The Data Science Process.

Unit IV	Data Wrangling	6 Hours

Combining and Merging Data Sets, Database style Data Frame Merges, Merging on Index, Concatenating Along an Axis, Combining Data with Overlap , Reshaping and Pivoting, Reshaping with Hierarchical Indexing, Data Transformation, Removing Duplicates, Replacing Values.

Unit V	Plotting and Visualization	6 Hours

A Brief matplotlib API Primer, Figures and Subplots, Colors, Markers, and Line Styles, Ticks, Labels, and Legends, Annotations and Drawing on a Subplot, Saving Plots to File,Plotting Functions in pandas, Line Plots, Bar Plots, Histograms and Density Plots, Scatter Plots.

Unit VI	Data Cleaning and Preprocessing	6 Hours

Data cleaning overview, Identifying and handling outliers, Handling inconsistent data, Parsing and formatting dates, Type conversion and data normalization, String manipulation techniques, Regular expressions for data cleaning, Data discretization and binning, Feature extraction from text and dates, Handling categorical variables using encoding techniques, Scaling and standardizing numeric data, Detecting and resolving duplicate entries.	
Texts Books:	
1.	Wes McKinney- Python for Data Analysis- O'REILLY, ISBN:978-1-449-31979-3, 1st edition, October 2012.
2.	Rachel Schutt & O'neil-Doing Data Science-O'REILLY, ISBN:978-1-449-35865-5, 1st edition, October 2013.
Reference Books:	
1	Joel Grus-Data Science from Scratch: First Principles with Python-O'Reilly Media, 2015
2	Matt Harrison-Learning the Pandas Library: Python Tools for Data Munging, Analysis, and Visualization - O'Reilly, 2016.
Supplementary Readings:	
https://www.coursera.org/learn/foundations-of-data-science https://www.mygreatlearning.com/academy/learn-for-free/courses/data-science-foundations	

Third Year B. Tech. (CSE) (Semester–V)							
Course Code	**01CSL303**		**Course Name**	**Machine Learning**			
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3	0	0	3		25	25	50

Pre-requisites:

Course Objectives:

01	Terminology of Machine learning
02	Supervised Machine Learning Algorithms
03	Unsupervised Machine Learning Algorithms

Course Outcomes:

At the end of the course, students will be able to

01	Explain machine learning concepts.
02	Analyze the Machine learning model.
03	Design solution using machine learning techniques.

Course Contents

Unit I	Introduction to Machine Learning	6 Hours

Definition, Terminology, Types of learning, Machine Learning Problem categories, Machine learning architecture, process, Lifecyle, Performance measures, tools and framework, data visualization.

Unit II	Regression	7 Hours

Simple regression – hypothesis, cost function, parameter learning with gradient descent, learning rate, Gradient Descent for linear regression, examples, simple regression in matrix form. Multivariate Linear Regression – multiple features, hypothesis functions, Gradient Descent for multiple variables, Feature scaling, polynomial regression

Unit III	Logistic regression & Neural Network	7 Hours

Definition, logistic regression – hypothesis representation, decision boundary, cost function, gradient descent for logistic regression. multiclass classification, Regularization - Overfitting & Underfitting, cost function, Regularized Linear Regression, Regularized Logistic Regression Neural Networks- Neuron representation and model, Hypothesis for neuron, cost function, solution of a problem using single neuron. Gradient descent for a neuron. Neural network, Multiclass classification with neural network. Learning in neural network-backpropagation algorithm

Unit IV	Support Vector Machine	6 Hours

Introduction to Support Vector Machine(SVM), Polynomial Support Vector Machines, Types of SVM Hyperplanes and Support Vectors, Working of SVM, Advantages and Disadvantages of SVM, Applications of SVM, Dot product, Use of Dot product in SVM, Margin, Hard Margin SVM, Soft margin SVM, SVM Kernels, Types of Kernels, Parameters in Kernelized SVC, Computing the SVM classifier

Unit V	Decision Tree, Naïve Bayes and K-Nearest	6 Hours

Decision trees: definition, terminology, the need, advantages, and limitations. constructing and understanding Decision trees, common problems with Decision trees, Decision tree algorithms, random forest, examples. Conditional probability and Naïve Bayes Classifier Instance-based classifier – K- Nearest Neighbor Classifier		
Unit VI	**Clustering and Association**	**6 Hours**
Clustering, K Means clustering, Hierarchical clustering, Association Rule mining		
Texts Books:		
1.	An Introduction to Statistical Learning: with applications in Python- Gareth James, Daniela Witten, Travor Hastie, Robert Tibshirani, Jonathan Taylor, Springer	
2.	Machine Learning: A Probabilistic Approach- Kevin Murphy, MIT Press, 1st Edition, 2012, ISBN No.: 978-0262-30616-4	
Reference Books:		
1	Discrete Mathematics - Semyour Lipschutz, MarcLipson (MGH), Schaum's outlines	
2	Probability and Statistics- John Schiller, Murray R. Spigel (MGH), Schaum's outlines 3. Probability	
Supplementary Readings:		
1. https://www.iith.ac.in/~aravind/Files-DM/LLM-MFCS-2004.pdf		

Third Year B. Tech. (CSE) (Semester–V)						
Course Code	**01CSP304**		**Course Name**	**Machine Learning Lab**		
Teaching Scheme					**Evaluation Scheme**	
L	T	P	Credits		CIE Marks	SEE Marks
--	--	2	1		50	50

Pre-requisites:

Course Objectives:

01	Design and implement supervised machine learning algorithm
02	Design and implement unsupervised machine learning algorithm

Course Outcomes:

At the end of the course, students will be able to

01	Describe the fundamentals of machine learning.
02	Design a model for classification or regression
03	Analyze machine learning model

Guidelines for Experiments

A minimum of 10 experiments will be performed on the topics listed below.

- Simple Linear Regression
- Multiple Linear Regression
- Logistic Regression.
- Multi-class Classification
- Neural Network
- K-means Clustering
- Association rule mining
- Naïve bay's classifier for text data
- Support Vector Mahine

Third Year B. Tech. (CSE) (Semester–V)							
Course Code	**01CSP305**	**Course Name**	**Database Engineering Lab**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		CIE Marks	SEE Marks	
--	--	2	1		50	50	

Pre-requisites: Data Structure

Course Objectives:

01	Introduce students to the setup and use of relational database management systems
02	Enable students to design logical data models using ER/EER diagrams and normalize data effectively.
03	Provide hands-on experience in writing SQL queries for data definition, manipulation, and control.
04	Develop skills in PL/SQL programming including procedures, functions, triggers, and cursors
05	Demonstrate real-time transaction management, concurrency control, and recovery operations.

Course Outcomes:

At the end of the course, students will be able to

01	Describe the fundamentals of database management systems, including ER modeling, normalization, and SQL standards
02	Design a relational database schema for real-world applications using ER diagrams and normalization techniques.
03	Analyze and optimize database queries using subqueries, joins, indexing, transactions, and query plans
04	Implement SQL and PL/SQL programs including DDL, DML, functions, procedures, cursors, and triggers

Guidelines for Experiments

A minimum of 10-12 experiments will be performed on the topics listed below.

- Installation and Demonstration of DBMS like MySql
- Draw E-R Diagram for different applications like – Library Management Systems, College Management Systems, Hospital Management Systems etc.
- Convert E-R Diagrams into relational tables and Normalize given relations up to BCN.
- Use DDL Statements to Crete, Alter, Drop, Rename, Truncate Tables
- Use DML Statements to Insert, Select, Update, Delete Data
- Use of aggregate functions, group by – having clause and order by clause.
- Use of Joins and Nested Queries
- Use of Set Operations
- Creation of Indices and Views in SQL
- Implement PL/SQL procedure and Function
- Implement PL/SQL Cursor.
- Implement Triggers in PL/SQL.
- Demonstration of Log based recovery.
- Study of concurrency control mechanisms
- Query Execution Plan and Optimization (EXPLAIN PLAN, Indexing)
- Design and implement a complete database system for a real-life application

Supplementary Readings:

- Oracle PL/SQL Programming – Steven Feuerstein
- Learning SQL – Alan Beaulieu
- Online Documentation (PostgreSQL, Oracle, MySQL)

Third Year B. Tech. (CSE) (Semester–V)						
Course Code	**01CSP334**		**Course Name**	**Introduction to Cloud Computing and Cloud Services Lab**		
Teaching Scheme					**Evaluation Scheme**	
L	T	P	Credits		CIE Marks	SEE Marks
--	--	2	1		50	--

Course Outcomes	
At the end of the course, students will be able to	
01	Demonstrate provision and manage virtual compute resources using AWS EC2, Auto Scaling, and Elastic Load Balancer
02	Implement and manage cloud-based storage solutions using AWS services such as S3, EBS, and EFS for various data access and lifecycle requirements.
03	Apply event-driven and secure computing principles to build scalable and secure cloud applications using AWS Lambda, IAM, and monitoring tools.

Guidelines for Experiments

A minimum of 8 experiments will be performed on the topics listed below.

- Launching an EC2 Instance
- Hosting a Static Website on Amazon S3
- Creating and Attaching EBS Volumes
- Auto Scaling and Load Balancing with EC2
- Backing Up and Restoring Data with Amazon S3 and Lifecycle Policies
- Using AWS Lambda to Trigger Functions on S3 Events
- Implementing IAM for Secure Compute and Storage Access
- Creating AMIs and Launching EC2 Instances from AMIs

Supplementary Readings:

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Third Year B. Tech. (CSE) (Semester–V)							
Course Code	**01CSP335**	**Course Name**	**Deep Learning Lab**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		CIE Marks	SEE Marks	
--	--	2	1		50	--	

Course Outcomes:	
At the end of the course, students will be able to	
01	Explain the foundational concepts of deep neural networks
02	Construct deep learning networks for given task.
03	Analyze different deep learning networks on various evaluation parameters

Guidelines for Experiments

A minimum of 8 experiments will be performed on the topics listed below.

- Installing and working with Tensorflow and Keras with Anaconda or Miniconda
- Introduction and working with Google Colab for using GPUs and TPUs for large projects
- Design and develop a model for predicting house prices.
- Design and develop shallow neural network for classifying hand written digit recognition.
- Design and develop deep neural network for classifying hand written digit recognition.
- Design and develop CNN for classifying images.
- Develop a CNN model with transfer learning for classifying images.
- Develop a CNN model to perform image segmentation.
- Develop a CNN model with residual block.
- Design and develop Xception-like model.
- Visualizing convnet filters.
- Identify sentiment of given text using unigram / bigram with TF-IDF encoding
- Design and develop RNN model for temperature forecasting
- Design and develop GRU/LSTM model for sentiment analysis.
- Hyper-parameters tuning for getting better result
- Design and develop a deep learning model for handling multiple input and multiple output
- Handling callbacks of deep neural network.
- Monitoring and visualization of model with tensorboard
- Design and develop a transformer based model for text classification.
- Design and develop a model for transferring source language to destination language.

Supplementary Readings:

1. https://developers.google.com/machine-learning

Third Year B. Tech. (CSE) (Semester–V)						
Course Code	**01CSP336**	**Course Name**	**Foundations of Data Science Lab**			
Teaching Scheme					**Evaluation Scheme**	
L	T	P	Credits		CIE Marks	SEE Marks
--	--	2	1		50	--

Course Outcomes:	
At the end of the course, students will be able to	
01	Perform various operations on numpy arrays
02	Importing data from different file formats using pandas
03	Draw different types of charts using matplotlib

Guidelines for Experiments

A minimum of 8 experiments will be performed on the topics listed below.

- Creating NumPy Array
- The Shape and Reshaping of NumPy Array
- Expanding and Squeezing a NumPy Array
- Indexing and Slicing of NumPy Array
- Stacking and Concatenating Numpy Arrays
- Perform operations using pandas
- Read the various file formats using pandas
- Demonstrate web scraping using python
- Preforming pre-processing techniques on dataset
- visualizations using matplotlib

Supplementary Readings:

https://www.coursera.org/learn/foundations-of-data-science

https://www.mygreatlearning.com/academy/learn-for-free/courses/data-science-foundations

Third Year B. Tech. (CSE) (Semester–V)							
Course Code	**01CSD351**			**Course Name**	**Software Mini Project**		
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		CIE Marks	SEE Marks	
0	0	4	2		50	50	

Pre-requisites: Software Engineering

Course Outcomes:

01	Create the requirement specifications and design document of their project
02	Design code according to the structure specified in SRS.
03	Perform unit as well as integrated testing.
04	Write a report covering details of the project.
05	Present the project work.

Project Guidelines

The group will continue to work on the project selected during the semester IV and submit the completed project work to the department at the end of semester V as mentioned below.

The project report in the spiral bound complete in all respect with the following : -

a) Problem specifications
b) System definition – requirement analysis.
c) System design – dataflow diagrams, database design
d) System implementation – algorithm, code documentation
e) Test results and test report.
f) In case of object oriented approach – appropriate process be followed.

CIE will be assessed by a project guide. SEE examination will be conducted by internal and external examiners as appointed by the COE.

B.Tech. CSE (Semester–V)							
Course Code	**01OEL303**			**Course Name**	**Startup and Entrepreneurship**		
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-Marks	SEE Marks
3			3		25	25	50

Course Objectives:	
01	To understand the foundational concepts and significance of entrepreneurship
02	To develop the ability to generate and recognize innovative business ideas
03	To learn how to create a sustainable and scalable business model.
04	To gain knowledge of legal and ethical considerations in starting a business.
05	To acquire skills in market research and competitor analysis.
06	To understand different funding sources and develop financial management strategies.
07	To enhance the ability to write compelling project proposals for innovative business ideas.

Course Outcomes:	
At the end of the course, students will be able to	
01	To cultivate an entrepreneurial mindset navigating the journey from idea inception to startup creation
02	To Develop a sustainable business model for a tech-based product or service.
03	To identify different funding sources and financial management
04	To write project proposals for innovative business ideas

Course Contents		
Unit I	**Introduction to Entrepreneurship**	**5 Hours**
Concept of Entrepreneurship, factors affecting the emergence of entrepreneurship, Types of Entrepreneurs. Entrepreneurial approach. Characteristic of successful entrepreneurs; Entrepreneurship process; Women Entrepreneurs, Social entrepreneurship, International Entrepreneurship, Technology Entrepreneurship. Entrepreneurial challenges. Difference Startup and Entrepreneurship		
Unit II	**Legal and Ethical Considerations**	**5 Hours**
Legal issues – Forming a business entity, considerations, and criteria, requirements for the formation of a Private/Public Limited Company, Intellectual Property Rights, Business Structures and Incorporation, Ethical Issues in Startups, Patent Registration, Guest Lecture: Legal Expert in Startup Law		
Unit III	**Ideation and Opportunity Recognition**	**5 Hours**
Techniques for Generating Ideas, Identifying Market Needs, Validating Ideas, Project: Create a comprehensive business model canvas with financial projection		
Unit IV	**Business Models and Planning**	**5 Hours**

Methods to Initiate Ventures, Business Plan- Advantages of business planning, and Perspectives in business plan preparation, business plan elements, Business Model Canvas; Value Proposition, Customer Segments, and Channels, Digital Business Model, Project: Develop a Business Model Canvas.		
UNIT V	**Market Research and Analysis**	**5 Hours**
Understanding Market Size and Trends, Competitor Analysis, Customer Discovery and Validation, Project: Conduct Market Research, Building the Team and Networking Guest Lecture		
Unit VI	**Institutions Supporting Entrepreneurs**	**5 Hours**
Introduction to small scale industry (SSI), Policies regarding SSI sector. Micro, Small and Medium Enterprises Development Act 2006. Concessions and Incentives, a brief overview of financial institutions in India: SIDBI, NABARD, IDBI, SIDCO, Presenting to Investors		

Texts Books:	
1.	Kumar Arya, Entrepreneurship: Creating and Leading an Entrepreneurial Organization, Pearson, India.
2.	His rich. Peters, Entrepreneurship: Starting, Developing and Managing a New Enterprise, Irwin. Department of Commerce, University of Delhi 14
3.	Startup Engineering, by Raman Bansal
Reference Books:	
1.	Allen, K.R., Launching New Ventures: An Entrepreneurial Approach, Cengage Learning. 2. Ramachandran, K., Entrepreneurship Development, Tata McGraw Hill, India. 3. Roy, Rajeev, Entrepreneurship, Oxford University Press.
2.	Just Aspire: Notes on Technology, Entrepreneurship and the Future by Ajai Chowdhury
3.	School To Startup, by Rohit Sinha
4.	The Startup Guidebook, by A Samanta
Supplementary Readings/ Material/ Online Courses:	
1.	https://ocw.mit.edu/courses/15-390-new-enterprises-spring-2013/
2.	https://nptel.ac.in/courses/110107094
3.	https://youtu.be/N3-FZn_iQFU

Third Year B. Tech. (CSE) (Semester–VI)							
Course Code			**01CSL306**	**Course Name**	**Cryptography and Network Security**		
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3	1	0	4		25	25	50

Pre-requisites:

Course Objectives:

01	To understand the basics of cryptography and some key encryption techniques.
02	To understand modern cryptosystems.
04	To understand network and Web security protocols.

Course Outcomes:

At the end of the course, students will be able to

01	Explain fundamental concepts in information security.
02	Apply cryptographic algorithms for information security
03	Analyze network security protocols and web security protocols.

Course Contents

Unit I	Overview and Classical Encryption Techniques	6 Hours

Overview: Computer Security Concepts, The OSI Security Architecture, Security Attacks, Security Services, Security Mechanisms, A Model for Network Security, Classical Encryption Techniques: Symmetric Cipher Model, Substitution, Techniques, Transposition Techniques, Rotor Machines

Unit II	Block Ciphers and Advanced Encryption Standard	6 Hours

Block Cipher Principles, The Data Encryption Standard (DES), A DES Example, The Strength of DES, Differential and Linear Cryptanalysis, Block Cipher Design Principles Finite Field Arithmetic, AES Structure, AES Transformation Functions, AES Key Expansion, An AES Example, AES Implementation.

Unit III	Public Key Cryptography	7 Hours

Principles of Public-Key Cryptosystems, The RSA Algorithm, Diffie-Hellman Key Exchange, ElGamal Cryptosystem.

Unit IV	Cryptographic Data Integrity Algorithms	7 Hours

Cryptographic Hash Functions: Applications of Cryptographic Hash Functions, Two Simple Hash Functions, Requirements and Security, Hash Functions Based on Cipher Block Chaining, Secure Hash Algorithm (SHA), SHA-3

Message Authentication Code: Message Authentication Requirements, Message Authentication Functions, Requirements for Message Authentication Codes, Security of MACs, MACs Based on Hash Functions: HMAC, MACs Based on Block Ciphers: DAA and CMAC, Authenticated Encryption: CCM and GCM

Digital Signatures: Elgamal Digital Signature Scheme, Schnorr Digital Signature Scheme, NIST Digital Signature Algorithm

Unit V	Mutual Trust and Key Management	6 Hours

Key Management and Distribution: Symmetric Key Distribution Using Symmetric Encryption, Symmetric Key Distribution Using Asymmetric Encryption, Distribution of Public Keys, X.509 Certificates, Public-Key Infrastructure

Unit VI	Network and Internet Security	7 Hours
IP Security: IP Security Overview, IP Security Policy, Encapsulating Security Payload, Combining Security Associations, Internet Key Exchange, Cryptographic Suites Transport-Level Security: Web Security Considerations, Secure Sockets Layer, Transport Layer Security, HTTPS Electronic Mail Security: Pretty Good Privacy, S/MIME, Domain Keys Identified Mail, Web Security		
Texts Books:		
1.	Williams Stallings – Cryptography and Network security principles and practices, Pearson Education (LPE), Sixth Edition	
2.	Cryptography and network security – Atul Kahate (TMGH)	
Reference Books:		
1	Handbook of Applied Cryptography - Menezes, A. J., P. C. Van Oorschot, and S. A. Vanstone	
Supplementary Readings:		
1.	https://blog.gigamon.com/2019/06/13/what-is-network-security-14-tools-and-techniques-to-know	

Third Year B. Tech. (CSE) (Semester–VI)							
Course Code	**01CSL337**	**Course Name**	**Networking and Security in Cloud**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3	0	0	3		25	25	50

Course Outcomes:

At the end of the course, students will be able to

01	Describe the basic concepts of AWS networking.
02	Design subnets in Amazon VPC
03	Implement DNS system in AWS Cloud

Course Contents

Unit I	Amazon VPC – Subnets, Route Tables, Internet Gateways	6 Hours

Introduction to Amazon VPC (Virtual Private Cloud), Subnets in Amazon VPC, Route Tables – Theory and Configuration, Internet Gateways (IGW), Putting it All Together – Building a Public and Private Network in VPC, Troubleshooting and Best Practices in VPC Networking

Unit II	Security Groups and Network ACLs	6 Hours

Introduction to Network Security in AWS, Deep Dive into Security Groups, Deep Dive into Network ACLs, Security Groups vs Network ACLs, Real-Time Configuration and Troubleshooting, Advanced Use Cases and Best Practices

Unit III	AWS Route S3 – DNS and Domain Management	6 Hours

Introduction to DNS and Route S3, Domain Registration and Management, Hosted Zones and DNS Records Management, Routing Policies in Route S3, Health Checks and DNS Failover, Route S3 Integration and Best Practices

Unit IV	AWS CloudFront – Content Delivery Network	6 Hours

Introduction to AWS CloudFront and CDN Basics, Architecture and Components of AWS CloudFront, Setting Up AWS CloudFront Distribution, Security Features in AWS CloudFront, Performance Optimization and Monitoring, Advanced Use Cases and Hands-on Demo

Unit V	AWS Certificate Manager and Encryption (KMS)	6 Hours

Introduction to AWS Certificate Manager (ACM), Managing SSL/TLS Certificates with ACM, Introduction to AWS Key Management Service (KMS), Encryption and Decryption using AWS KMS, Advanced Key Management and Security Controls, Real-world Use Cases and Best Practices

Unit VI	Identity Federation and Multi-Factor Authentication (MFA)	6Hours

Introduction to Identity Federation, Identity Federation Standards and Protocols, Federation Architectures and Deployment Models, Introduction to Multi-Factor Authentication (MFA), MFA Technologies and Implementation, Security Considerations, Challenges, and Future Trends

Texts Books:

1.	AWS Certified Advanced Networking – Specialty Study Guide: Specialty (ANS-C01) Author: Todd Montgomery, Tim Szigeti, and multiple contributors Publisher: Wiley

Reference Books:

1	AWS for Solutions Architects: Design and Manage Powerful Cloud Solutions

	Author: Alberto Artasanchez, Ramon Resma Publisher: Packt Publishing
Supplementary Readings:	
https://docs.aws.amazon.com/whitepapers/latest/aws-overview/introduction.html	

Third Year B. Tech. (CSE) (Semester–V)							
Course Code	**01CSL338**	**Course Name**	**Generative AI**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3	0	0	3		25	25	50

Pre-requisites:

Course Objectives:

01	Generative AI with Deep Learning
02	Variational Autoencoders, Generative Adversarial Network
03	Application and future of Generative AI

Course Outcomes:

At the end of the course, students will be able to

01	Explain the foundational concepts of deep neural networks and generative modeling.
02	Apply deep learning and generative modeling techniques to solve problems.
03	Analyze the performance and limitations of various generative models.

Course Contents

Unit I	Generative Modeling	6 Hours

Definition, Generative Versus Discriminative Modeling, Advances in Machine Learning, The Rise of Generative Modeling, The Generative Modeling Framework, Probabilistic Generative Models, Hello World! program, First Probabilistic Generative Model, Naive Bayes, Hello World! Continued, The Challenges of Generative Modeling, Representation Learning

Unit II	Deep Learning	7 Hours

Structured and Unstructured Data, Deep Neural Networks, Keras and TensorFlow, Your First Deep Neural Network, Loading the Data, Building the Model, Compiling the Model, Training the Model, Evaluating the Model, Improving the Model, Convolutional Layers, Batch Normalization, Dropout Layers.

Unit III	Variational Autoencoders	7 Hours

The Art Exhibition, Autoencoders, First Autoencoder, The Encoder, The Decoder, Joining the Encoder to the Decoder, Analysis of the Autoencoder, The Variational Art Exhibition, Building a Variational Autoencoder, The Encoder, The Loss Function, Analysis of the Variational Autoencoder, Using VAEs to Generate Faces, Training the VAE, Analysis of the VAE, Generating New Faces, Latent Space Arithmetic, Morphing Between Faces.

Unit IV	Generative Adversarial Networks	6 Hours

Ganimals, Introduction to GANs, Your First GAN, The Discriminator, The Generator, Training the GAN, GAN Challenges, Oscillating Loss, Mode Collapse, Uninformative Loss, Hyperparameters, Tackling the GAN Challenges, Wasserstein GAN, Wasserstein Loss, The Lipschitz Constraint, Weight Clipping, Training the WGAN, Analysis of the WGAN, WGAN-GP, The Gradient Penalty Loss, Analysis of WGAN-GP

Unit V	Application of GenAI	6 Hours

CycleGAN, Your First CycleGAN, Overview, The Generators (U-Net), The Discriminators, Compiling the CycleGAN, Training the CycleGAN, Analysis of the CycleGAN, Creating a CycleGAN to Paint Like Monet,

The Generators (ResNet), Analysis of the CycleGAN, Neural Style Transfer, Content Loss, Style Loss, Total Variance Loss, Running the Neural Style Transfer, Analysis of the Neural Style Transfer Model.		
Unit VI	**Future of Generative AI**	**6 Hours**
Five Years of Progress, The Transformer, Positional Encoding, Multihead Attention, The Decoder, Analysis of the Transformer, BERT, GPT-2, MuseNet, Advances in Image Generation, ProGAN, Self-Attention GAN (SAGAN), BigGAN, StyleGAN, Applications of Generative Modeling, AI Art, AI Music.		
Texts Books:		
1.	Generative Deep Learning Teaching Machines to Paint, Write, Compose, and Play - David Foster - O'reilly	
2.	Deep Learning for Vision Systems - Mohamed Elgendy - Manning	
Reference Books:		
1	GANs in Action: Deep learning with Generative Adversarial Networks - Jakub Langr, Vladimir Bok - Manning	
Supplementary Readings:		
1. https://github.com/davidADSP/Generative_Deep_Learning_2nd_Edition/tree/main		

Third Year B. Tech. (CSE) (Semester–VI)							
Course Code	**01CSL339**	**Course Name**	**Data Warehousing and Business Intelligence**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3	0	0	3		25	25	50

Course Objectives:	
01	Inculcate Conceptual, Logical, and Physical design of Data Warehouses OLAP applications and OLAP deployment
02	Understand a data warehouse or data mart to present information needed by management in a form that is usable

Course Outcomes:	
At the end of the course, students will be able to	
01	Describe components and working of Data warehouse and Business Intelligence systems
02	Explain dimensional modelling using fact table and dimension table
03	Explain various components and working of ETL Systems
04	Explain analytical tools and reporting tools for Business Data Analysis

Course Contents		
Unit I	**Introducing the Technical Architecture**	**06 Hours**
The value of architecture, Technical Architecture overview, Back room Architecture, Presentation Server Architecture, Front room Architecture, Infrastructure, Metadata, Security.		
Unit II	**Introducing Dimensional Modelling**	**06 Hours**
Making the Case for Dimensional Modelling, Dimensional Modelling primer, Enterprise Data Warehouse Bus Architecture, More on Dimensions & Facts.		
Unit III	**Designing the Dimensional Modelling**	**08 Hours**
Modelling Process overview, Getting Organized, Four Step Modelling Process, Design the Dimensional Model.		
Unit IV	**Introducing Extract, Transformation & Load**	**06 Hours**
Round up the requirements, the 34subsystems of ETL, Extracting Data, Cleaning & Conforming data, Delivering Data for Presentation		
Unit V	**Introducing Business Intelligence Applications**	**08 Hours**
Use Cases Importance of B.I. Applications, Analytical cycle for B.I, Types of B.I. Applications, Navigating Applications via the B.I portal		
Unit VI	**Designing & Developing B.I Applications**	**06 Hours**
B.I. Application resource planning, B.I. Application Specification, B.I. Application Development, B.I. Application maintenance		

Texts Books:	
1.	Ralph Kimball, "The Data Warehouse Lifecycle toolkit', 2nd edition, Wiley India
Reference Books:	

1	Data Mining Techniques and Applications: An Introduction, Hongbo Du, Cengage Learning
2	Data Mining: VikramPudi and P. Radha Krishna, Oxford Publisher.
Supplementary Readings:	
https://www.geeksforgeeks.org/dbms/data-warehousing/	

Third Year B. Tech. (CSE) (Semester–VI)						
Course Code	**01CSP340**		**Course Name**	**Networking and Security in Cloud Lab**		
Teaching Scheme					**Evaluation Scheme**	
L	T	P	Credits		CIE Marks	SEE Marks
--	--	2	1		50	--

Course Outcomes:	
At the end of the course, students will be able to	
01	Design and implement secure and scalable network architectures in AWS
02	Apply AWS security services and access management tools to protect cloud resources.
03	Demonstrate the ability to monitor, protect, and troubleshoot cloud applications using VPN connectivity

Guidelines for Experiments

A minimum of 8 experiments will be performed on the topics listed below.

- VPC Creation and Configuration
- Security Groups and NACLs
- Launch EC2 in Private Subnet with NAT Gateway
- Configure a VPN Connection
- IAM Roles and Policies for EC2
- Setup a Bastion Host for Secure SSH
- S3 Bucket Security Configuration
- AWS CloudTrail and CloudWatch for Security Monitoring

Supplementary Readings:

1. https://www.w3schools.com/dsa/index.php

Third Year B. Tech. (CSE) (Semester–V)							
Course Code	**01CSP341**		**Course Name**	**Generative AI Lab**			
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		CIE Marks	SEE Marks	
--	--	2	1		50	--	

Course Outcomes:	
At the end of the course, students will be able to	
01	Explain the foundational concepts of deep neural networks, generative modeling
02	Build and train generative deep learning models for real-world data generation.
03	Analyze the performance and limitations of various generative models.

Guidelines for Experiments

A minimum of 8 experiments will be performed on the topics listed below.

- Next word prediction
- Naive Bayes Classifier as a Probabilistic Generative Model
- Visualize Latent Space in a Variational Autoencoder
- Build a Basic GAN to Generate Handwritten Digits
- Compare Loss Patterns in GAN vs. Wasserstein GAN
- Neural Style Transfer Using Pretrained VGG
- CycleGAN for Image Translation

Supplementary Readings:

1. https://developers.google.com/machine-learning/gan/generative

Third Year B. Tech. (CSE) (Semester–IV)						
Course Code	**01CSP342**	**Course Name**	**Data Warehousing and Business Intelligence Lab**			
Teaching Scheme					**Evaluation Scheme**	
L	T	P	Credits		CIE Marks	SEE Marks
--	--	2	1		50	50

Pre-requisites:

Course Objectives:

01	Inculcate Conceptual, Logical, and Physical design of Data Warehouses OLAP applications and OLAP deployment
02	Design a data warehouse or data mart to present information needed by management in a form that is usable
03	Emphasize hands-on experience working with all real data sets.
04	Test real data sets using popular data mining tools such as WEKA, Python Libraries
05	Inculcate Conceptual, Logical, and Physical design of Data Warehouses OLAP applications and OLAP deployment

Course Outcomes:

At the end of the course, students will be able to

01	Install and configure BI Systems.
02	Design dimensional modelling using fact table and dimension table.
03	Design ETL process
04	Apply analytical and reporting tools for various BI tasks.

Guidelines for Experiments

A minimum of 8 to 10 experiments will be performed on the topics listed below.

- Install and configure BI Systems
- Fact and dimension tables for specific applications
- Dimensional Modeling and Schema Design for given business process
- Data profiling and pre-processing of data before Extract Transfer Load
- Data Cleansing and Deduplication using suitable tools
- ETL using suitable ETL tools
- Slowly changing dimensions using suitable tools
- Predictive Analytics and Data Mining
- Data Visualization and Reporting
- Data Governance and Metadata Management
- Dashboards using suitable tools
- Scorecards using suitable tools

Supplementary Readings:

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Third Year B. Tech. (CSE) (Semester–VII)							
Course Code		**01CSL343**	**Course Name**	**Cloud Database and Analytics Services**			
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3	0	0	3		25	25	50

01	Understand core concepts, architecture, and functionalities of AWS-managed database and data analytics services
02	Introduce cloud-native relational and NoSQL databases
03	Understand performance optimization, scalability, security, and cost-efficiency strategies across AWS data services

Course Outcomes:

At the end of the course, students will be able to

01	Describe the basic concepts of AWS database.
02	Explain the basic concepts of AWS ETL & Data Lakes
03	Describe the basic concepts of data visualization.

Course Contents

Unit I	Amazon RDS – Relational Database Service	6 Hours
Introduction to Amazon RDS, Setting Up an Amazon RDS Instance, Managing and Operating Amazon RDS, High Availability and Scalability in RDS, Security and Compliance in Amazon RDS, Advanced Features and Cost Optimization		
Unit II	**Amazon DynamoDB – NoSQL Database**	**6 Hours**
Introduction to NoSQL and DynamoDB Basics, Data Modeling in DynamoDB, DynamoDB Operations and APIs, Performance, Scalability, and Capacity Management, Security, Backup, and Monitoring, Advanced Topics and Use Cases		
Unit III	**Amazon Redshift – Data Warehousing**	**6 Hours**
Introduction to Data Warehousing & Amazon Redshift Overview, Amazon Redshift Architecture & Components, Setting Up & Managing Amazon Redshift Clusters, Loading & Querying Data in Amazon Redshift, Performance Tuning and Optimization in Redshift, Integration, Monitoring, and Best Practices		
Unit IV	**Amazon Aurora – High-Performance DB**	**6 Hours**
Introduction to Amazon Aurora, Aurora Storage and Replication Architecture, Performance and Scalability of Aurora, Security and Compliance in Amazon Aurora, Managing and Operating Aurora Databases, Advanced Features and Use Cases		
Unit V	**AWS Glue and AWS Lake Formation – ETL & Data Lakes**	**6 Hours**
Introduction to Data Lakes and ETL Concepts, Introduction to AWS Glue, AWS Glue ETL Jobs and Development, Introduction to AWS Lake Formation, Security, Access Control, and Governance in Lake Formation, Building and Managing Data Lakes using AWS Glue & Lake Formation		
Unit VI	**Amazon QuickSight – Data Visualization**	**6Hours**
Introduction to Amazon QuickSight, Getting Started with Amazon QuickSight, Building Visualizations in QuickSight, Advanced Data Preparation and Analysis, Sharing and Collaboration, Hands-on Project & Best Practices		

Texts Books:

1.	Amazon Web Services in Action (Third Edition)- Michael Wittig & Andreas Wittig- Manning Publications
2.	Learning Amazon Web Services (AWS): A Hands-On Guide to the Fundamentals of AWS Cloud- Mark Wilkins-Pearson Education
Reference Books:	
1	Discrete Mathematics - Semyour Lipschutz, MarcLipson (MGH), Schaum's outlines Probability and Statistics- John Schiller, Murray R. Spigel (MGH), Schaum's outlines 3. Probability
Supplementary Readings:	
https://aws.amazon.com/training/	

Third Year B. Tech. (CSE) (Semester–VI)							
Course Code	**01CSL344**	**Course Name**	**Large Language Models**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3	0	0	3		25	25	50

Pre-requisites:

Course Objectives:

01	Transformers and its architectures
02	Introduce LLM Models
03	Understand Embedding Space

Course Outcomes:

At the end of the course, students will be able to

01	Describe architecture of Transformers
02	Describe basic concepts of Large Language Models
03	Analyze embedding space

Course Contents

Unit I	Understanding Language Models	6 Hours
Language AI, History of Language AI, Definition of LLM, Limited Resources, Interfacing Large Language Model, Generating Text		
Unit II	**Tokens and Embeddings**	**7 Hours**
LLM Tokenization, Token Embeddings, Text Embeddings, Word Embeddings beyond LLMs, Embeddings for recommendation system,		
Unit III	**Transformer Model**	**7 Hours**
An Overview of Transformer Models, Recent Improvements to the Transformer Architecture,		
Unit IV	**Text Classification**	**6 Hours**
The Sentiment of Movie Reviews, Text Classification with Representation Models, Model Selection, Using a Task-Specific Model, Classification Tasks That Leverage Embedding, Text Classification with Generative Model,		
Unit V	**Text Clustering and Topic Modeling**	**6 Hours**
A Common Pipeline for Text Clustering, From Text Clustering to Topic Modeling,		
Unit VI	**Prompt Engineering**	**6 Hours**
Using Text Generation Model, Intro to Prompt Engineering, Advanced Prompt Engineering, Chain Prompting, Reasoning with Generative Models, Output Verification		

Texts Books:

1.	Build a Large Language Model (From Scratch)- Sebastian Raschka - MEAP
2.	Hands on Large Langauge Models- Jay Alammar, Maarten Grootendorst, Oreilly

Reference Books:

1	Building LLMs for Production: Enhancing LLM Abilities and Reliability with Prompting, Fine-Tuning, and RAG- Louis-François Bouchard, Louie Peters-Kindle Edition

Supplementary Readings:	
1. https://github.com/rasbt/LLMs-from-scratch/tree/main/ch01	

Third Year B. Tech. (CSE) (Semester–VI)							
Course Code	**01CSL345**	**Course Name**	**Data Analytics**				
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
3	0	0	3		25	25	50

Pre-requisites:

Course Objectives:

01	To introduce key concepts, tools, and techniques of data analytics.
02	To enable students to analyze real-world data using descriptive, predictive, and prescriptive analytics.
03	To build foundational knowledge for advanced subjects like Social Network Analysis and Time Series.

Course Outcomes:

At the end of the course, students will be able to

01	Understand data analytics processes, types, and technologies
02	Apply statistical, exploratory, and predictive analytics to datasets
03	Evaluate analytics models and interpret insights for decision-making

Course Contents

Unit I	Introduction to Data Analytics	6 Hours
Data Analytics: Definition, Importance, and Applications, Types of Analytics: Descriptive, Predictive, and Prescriptive, Analytics Lifecycle: Business Understanding, Data Acquisition, Modeling, Deployment, Overview of Analytics Tools: Python, R, Tableau, Power BI		
Unit II	**Exploratory Data Analysis and Preprocessing**	**6 Hours**
Data Types, Summary Statistics, Correlation, Visualization: Histograms, Boxplots, Pairplots, Heatmaps, Data Cleaning: Handling missing values, duplicates, outliers, Feature Engineering: Encoding, Normalization, Scaling		
Unit III	**Statistical Methods for Data Analytics**	**6 Hours**
Probability Distributions: Normal, Binomial, Poisson, Descriptive vs Inferential Statistics, Hypothesis Testing: t-test, ANOVA, Chi-square, Central Limit Theorem and Confidence Intervals		
Unit IV	**Predictive Modeling**	**6 Hours**
Supervised Learning: Concept and Workflow, Linear Regression, Logistic Regression, Decision Trees, Random Forest Classifiers, Model Evaluation: Confusion Matrix, ROC Curve, Cross-validation		
Unit V	**Unsupervised Learning and Feature Reduction**	**6 Hours**
Clustering Techniques: K-Means, DBSCAN, Hierarchical Clustering, Distance Metrics: Euclidean, Manhattan, Cosine Similarity, Principal Component Analysis (PCA), Feature Selection and Feature Importance		
Unit VI	**Big Data Tools and Case Studies**	**6 Hours**
Introduction to Big Data Analytics: Volume, Variety, Velocity, Hadoop and HDFS Basics, MapReduce Concepts, Spark Overview and PySpark for Data Analysis, Case Studies: Healthcare, Retail, Finance Applications		

Texts Books:

1.	Data Science for Business: What You Need to Know about Data Mining and Data-Analytic Thinking, Foster Provost & Tom Fawcett, O'Reilly Media, ISBN: 978-1449361327

2.	Data Science and Analytics, Vasant Dhar, Pearson Education, ISBN: 978-9332570313
3.	An Introduction to Statistical Learning: With Applications in R, James, Witten, Hastie, Tibshirani, Springer, ISBN: 978-1461471370
Reference Books:	
1	Data Science from Scratch: First Principles with Python, Joel Grus, O'Reilly Media, ISBN: 978-1492041139.
2	Advanced and Multivariate Statistical Methods: Practical Application and Interpretation, P. C. Mertler & R. A. Vannatta, Routledge, ISBN: 978-1138892286
3	Data Analytics, Anil Maheshwari, McGraw Hill Education, ISBN: 978-9353164979
Supplementary Readings:	
	https://www.kdnuggets.com https://towardsdatascience.com

Third Year B. Tech. (CSE) (Semester–VI)							
Course Code	**01CSP357**		**Course Name**	**Robotics Process Automation**			
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		CIE Marks	SEE Marks	
0	0	2	1		50	50	

Pre-requisites:

Course Objectives:

01	Understand the Fundamentals of RPA and Its Industry Applications
02	Gain Hands-on Experience with Leading RPA Tools
03	Design and Deploy Automation for Real-World Tasks
04	Build Job-Ready Skills with Projects and Certification Guidance

Course Outcomes:

At the end of the course, students will be able to

01	Describe basic concepts of RPA
02	Apply different tools used for RPA
03	Analyze different methodologies used in RPA

Course Contents

- Install and set up UiPath Community Edition and explore the interface (Studio, Orchestrator, Robot).
- Create a basic automation workflow using UiPath to perform arithmetic operations.
- Build a process that uses string operations and regular expressions for text extraction and validation.
- Automate reading and writing data from Excel files using Read Range, Write Range, and Write Cell activities.
- Filter data from an Excel DataTable and save the filtered results to a new file.
- Design a workflow to send automated emails with attachments using Outlook or Gmail.
- Create a bot that reads unread emails, saves attachments, and moves them to specific folders.
- Implement file operations: create, move, delete, and compress folders/files.
- Use UiPath to record basic desktop automation tasks like opening Notepad, typing, and saving a file.
- Automate a web data extraction task from an e-commerce or news website using Data Scraping.
- Create a workflow that interacts with web forms: fill input fields, click buttons, and handle pop-ups.
- Use selectors and UI elements to interact with dynamic web components.
- Automate a monthly sales report generation in Excel using input templates and formatting.
- Design a bot to auto-reply to emails and log responses in an Excel sheet.
- Use hotkeys and delay activities to handle application load times and keyboard shortcuts.

Texts Books:

1	Robotic Process Automation Handbook-Tom Taulli

Reference Books:

1	Essentials of Robotics Process Automation- Khanna publishing house

Supplementary Readings:

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Third Year B. Tech. (CSE) (Semester–VI)							
Course Code	**01CSP307**		**Course Name**	**Full Stack Web Development**			
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		CIE Marks	SEE Marks	
2	0	2	3		50	50	

Pre-requisites:

Course Objectives:

01	Introduce scripting language
02	Introduce front end framework for developing web pages
03	Creating web pages for given specification

Course Outcomes:

At the end of the course, students will be able to

01	Describe basic concepts of web application development
02	Design web application for the requirement employing JavaScript.
03	Design web application for the requirement employing React.

Course Contents

Unit I	JavaScript Fundamentals	3 Hours
Variables, data types, and operators, Functions, scope, and closures, Arrays, objects, and loops, ES6+ features: let/const, arrow functions, spread/rest, destructuring, Promises, async/await, and callbacks		
Unit II	**Introduction to ReactJS**	**4 Hours**
Introduction to ReactJS, features and advantages of React, single page applications, React vs traditional JavaScript frameworks; setting up React environment using Node.js, npm, Create React App / Vite, JSX syntax, expressions and attributes; functional components; props and data passing; state management using useState hook; event handling in React; conditional rendering; rendering lists and keys.		
Unit III	**Advanced ReactJS**	**4 Hours**
React hooks – useEffect and introduction to useContext; component lifecycle using hooks; forms in React and controlled components; lifting state up; routing using React Router – BrowserRouter, Routes, Route, Link; styling approaches in React; basic API integration using fetch or axios; handling loading and error states; project structure and best practices.		
Unit IV	**Node.js Essentials**	**5 Hours**
Introduction to Node.js and its architecture, Node.js runtime and event-driven programming, File system, buffers, streams, Creating a basic server with Node.js, Using npm and managing dependencies		
Unit V	**Express.js Framework**	**4 Hours**
Introduction to Express.js, Setting up RESTful routes (GET, POST, PUT, DELETE), Route parameters and query strings, Middleware and error handling, Creating a modular Express app		
Unit VI	**MongoDB and Mongoose**	**4 Hours**
Introduction to NoSQL and MongoDB, MongoDB CRUD operations, Mongoose ODM: schemas and models, Data validation and relationships, Connecting Express to MongoDB (local and Atlas)		

Texts Books:

1	Learning React- Alex Banks, Eve Porcello-Oreilly

Reference Books:

1	React Quick Start- Lionel Lopez

Supplementary Readings:
• https://www.w3schools.com/REACT/default.asp

Third Year B. Tech. (CSE) (Semester–VI)						
Course Code	**01CSP346**		**Course Name**	**Cloud Database and Analytics Services Lab**		
Teaching Scheme					**Evaluation Scheme**	
L	T	P	Credits		CIE Marks	SEE Marks
--	--	2	1		50	--

Course Objectives:	
01	Understand the concepts of AWS cloud-based data services
02	Introduce data lakes using Amazon RDS, DynamoDB, Redshift, Aurora, Glue, Lake Formation, and QuickSight.

Course Outcomes:	
At the end of the course, students will be able to	
01	Implement and manage relational and NoSQL databases using AWS services.
02	Develop and monitor ETL workflows and data lakes using AWS Glue and Lake Formation
03	Build interactive dashboards and visualizations using Amazon QuickSight for real-time business insights.

Guidelines for Experiments

A minimum of 8 experiments will be performed on the topics listed below.

- Set up and configure an Amazon RDS instance with MySQL or PostgreSQL.
- Perform backup, restore, and snapshot operations in Amazon RDS.
- Implement read replicas and multi-AZ deployment for high availability in RDS.
- Create and manage a DynamoDB table with partition and sort keys.
- Perform CRUD operations on DynamoDB using AWS SDK or CLI.
- Model and analyze access patterns in DynamoDB for optimized performance.
- Launch and configure an Amazon Redshift cluster.
- Load structured data into Redshift from S3 and query using SQL.
- Implement performance tuning and workload management in Redshift.
- Launch and operate Amazon Aurora with replication and failover.
- Use AWS Glue to create ETL jobs to transform and move data between services.
- Build and secure a data lake using AWS Lake Formation.
- Perform role-based access control and governance in Lake Formation.
- Connect QuickSight to AWS data sources and create interactive dashboards.
- Perform advanced data preparation and sharing in Amazon QuickSight

Supplementary Readings:

1. https://www.youtube.com/watch?v=9UHcPQfmZX8

Third Year B. Tech. (CSE) (Semester–VI)							
Course Code	**01CSP347**		**Course Name**	**Large Language Models Lab**			
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		CIE Marks	SEE Marks	
--	--	2	1		50	--	

Course Objectives:	
01	Introduce foundational principles of language models, including tokenization, embeddings, and transformer architecture.
02	Understand text classification, clustering, and generation tasks using pre-trained and fine-tuned large language models.
03	Understand prompt engineering and reasoning-based experiments with generative models.
Course Outcomes:	
At the end of the course, students will be able to	
01	Explain the basic concepts of large language models.
02	Use the large language models to solve real life problem.
03	Evaluate the prompts used for large language model.

Guidelines for Experiments

A minimum of 8 experiments will be performed on the topics listed below.

- Explore Tokenization with LLMs
- Generate Text using a Pre-trained Language Model
- Visualize and Compare Embeddings
- Build a Transformer from Scratch (Mini Version)
- Perform Text Classification using a Fine-Tuned Model
- Clustering News Articles using Sentence Embeddings
- Topic Modeling using BERTopic or LDA
- Design a Prompt to Solve a Task using GPT
- Experiment with Chain-of-Thought Prompting for Logical Reasoning Tasks
- Evaluate the Output of an LLM for Hallucination and Bias

Supplementary Readings:

1. https://onlinecourses.nptel.ac.in/noc25_cs161/preview

Third Year B. Tech. (CSE) (Semester–VI)						
Course Code	**01CSP348**		**Course Name**	**Data Analytics Lab**		
Teaching Scheme					**Evaluation Scheme**	
L	T	P	Credits		CIE Marks	SEE Marks
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Course Objectives:	
01	Introduce concepts, tools, and techniques used in data analytics, including types of analytics, data preprocessing, and statistical methods.
02	Understand data visualization, predictive modeling, and clustering techniques to explore and interpret real-world datasets.
03	Understand effectiveness of different analytics models and big data tools in solving domain-specific problems through case studies and evaluations.

Course Outcomes:	
At the end of the course, students will be able to	
01	Describe the data analytics lifecycle and perform data cleaning, transformation, and visualization using appropriate tools and techniques.
02	Develop and evaluate predictive and unsupervised learning models using statistical and machine learning methods.
03	Analyze complex datasets and assess the applicability of models in various domains.

Guidelines for Experiments

A minimum of 8 experiments will be performed on the topics listed below.

- Perform descriptive, predictive, and prescriptive analytics on sample datasets
- Use Python/R/Tableau/Power BI for basic data analysis and visualization
- Conduct exploratory data analysis (EDA) using histograms, boxplots, pairplots, and heatmaps
- Handle missing values, outliers, and duplicates in a dataset
- Apply feature engineering techniques: encoding, normalization, and scaling
- Visualize and analyze different probability distributions (Normal, Binomial, Poisson)
- Perform hypothesis testing: t-test, ANOVA, and Chi-square test on real-world data
- Demonstrate the Central Limit Theorem and calculate confidence intervals
- Build and evaluate supervised learning models: Linear Regression, Logistic Regression, Decision Trees, Random Forest
- Use confusion matrix, ROC curve, and cross-validation for model evaluation
- Apply clustering techniques: K-Means, DBSCAN, Hierarchical Clustering
- Compare distance metrics: Euclidean, Manhattan, and Cosine similarity
- Perform feature reduction using Principal Component Analysis (PCA)
- Use feature selection methods and determine feature importance
- Execute basic big data analysis using Hadoop and HDFS commands
- Perform data analysis using PySpark

Supplementary Readings:

1. https://grow.google/intl/en_in/data-analytics-course/

Third Year B. Tech. (CSE) (Semester–VI)							
Course Code	**01CSD352**		**Course Name**	**Project-I**			
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		CIE	SEE Marks	
0	0	4	2		50	50	

Pre-requisites: Software Engineering

Course Outcomes:

01	Define problem statement for selected project topic.
02	Analyze the requirements of the selected project topic.
03	Develop the design for providing solution to the selected problem statement.
04	Write a report covering SRS and design of the project
05	Give presentation on project.

Project Guidelines

Project group Formation:

- In order to form a project group, students are required to assemble team comprising four to five individuals.

Area of interest submission and guide allocation:

- Each group must submit their area of interest, along with the relevant student information, using the provided form. Subsequently, based on the indicated area of interest and the expertise of the faculty, each group will be assigned a guide.

Finalization of topic:

- It is necessary for the students to consult with their respective guides in order to select a suitable project topic.
- The finalization of the chosen topic will be conducted by the project evaluation team.

Synopsys preparation presentation(I):

- Once the topic has been selected and finalized, the group must proceed with the preparation and presentation of the project's synopsis. This presentation will serve as the first presentation of the project-I.
- The synopsis should encompass an introduction, related work, project topic selection including the problem statement, objectives, and methodology.

Presentation(II)

- In the second presentation, the group is required to develop a software requirement specification (SRS) document.
- The evaluation of the SRS will be based on various criteria, including requirement gathering, analysis, teamwork, and presentation skills.
- Furthermore, during the internal exam, the evaluation will be focused on the design and report of the project.
- Evaluation of internal exam is based on various criteria like tools, environmental factors, ethics, cost, self-learning, presentation, and team work.

Third Year B. Tech. (CSE) (Semester–VI)							
Course Code	**01OEL351**		**Course Name**	**German Language**			
Teaching Scheme					**Evaluation Scheme**		
L	T	P	Credits		SE-I Marks	SE-II Marks	SEE Marks
2	0	0	2		25	25	50

Pre-requisites:

Course Objectives:

01	Understand the basics of German Language and Pronunciations.
02	Understand grammar and make sentences.
03	Teach to read German Language.
04	Teach to write Letter and Essays.

Course Outcomes:

At the end of the course, students will be able to

01	Pronounce days, Months, Alphabets and Time.
02	Will be able to conduct daily conversation in German Language.
03	Read the German Language.
04	Write Letters and Essays in German Language.

Course Contents

Unit I	**Basic Introduction**	**5 Hours**
Alphabets, Numbers, days, Months, Time and Pronounsations.		
Unit II	**Grammar**	**5 Hours**
Verbs, Nouns, Plural and Prepositions.		
Unit III	**Family**	**7 Hours**
Making sentences, Talk about family and family members and progressive articles.		
Unit IV	**Reading**	**6 Hours**
Reading conversations at Restaurant, Railway station, Bus Stand, Vegetable market and Super markets.		
Unit V	**Translation**	**7 Hours**
Translation of German Language into English Language and English into German Language.		
Unit VI	**Writing Letter and Essay**	
Practice of Writing Letters, Writing Applications and Essays.		

Texts Books:

1.	Netzwerk Neu – AI Kursbuch and Ubungsbuch
2.	Sprachkurs Deutsch by Ulrich Haussermann

Reference Books:

1	Auf Deutsch Gesagt by Radio Rudolf Schneider

Supplementary Readings:

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